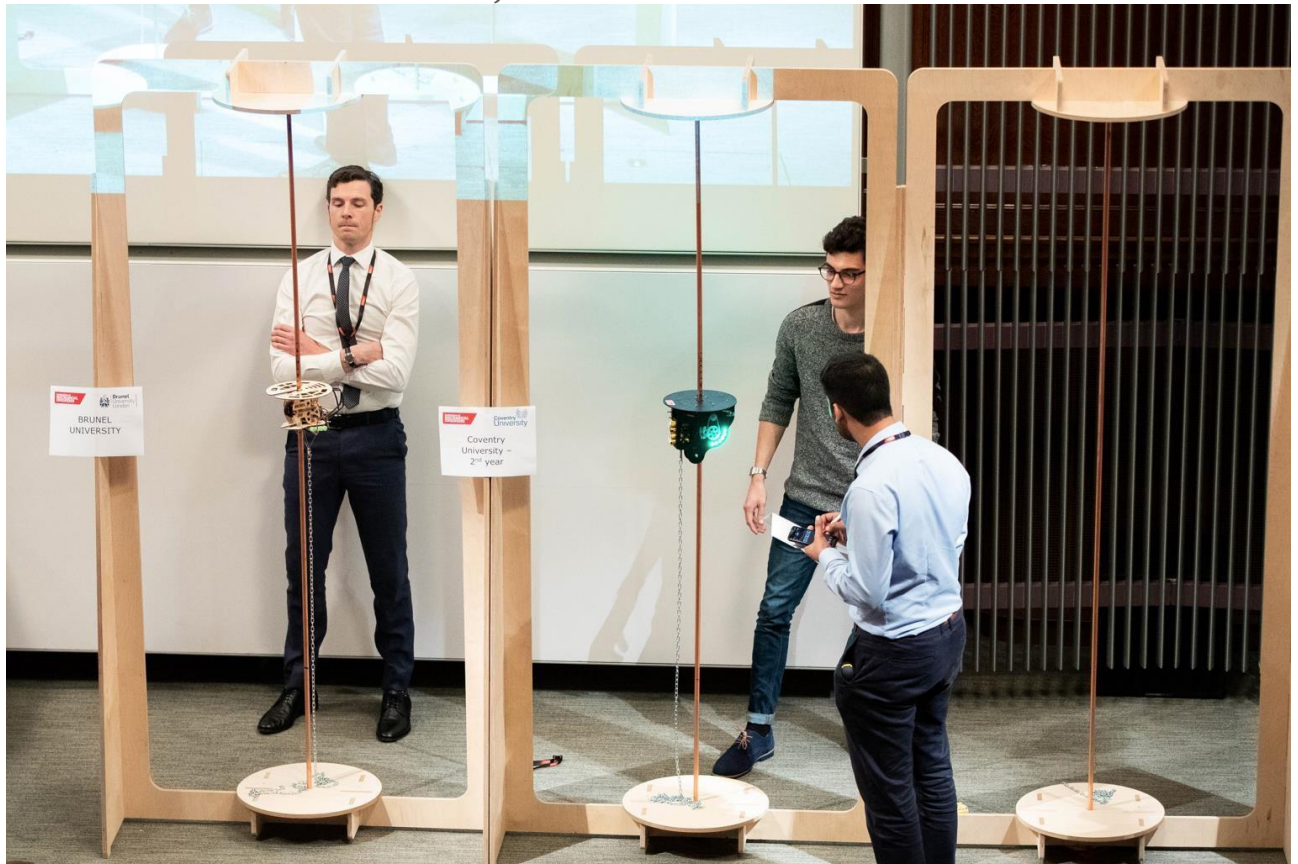


DESIGN CHALLENGE PROJECT SPECIFICATION AND RULES.

Institution of
**MECHANICAL
ENGINEERS**

EXTERNAL PIPE CLIMBER PROJECT - 2026



Project Specification and Rules for the 2026 External Pipe Climber Design Challenge

*Note: This Specification must be read in
conjunction with the General Specification 2026*

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1. Introduction

The purpose of the Design Challenge competition is to simulate a real-world project, allowing students to develop cross-functional skills. Working in teams, students will design, build and test a self-contained device from a precise Project Specification. The competition is open to teams of **two to five** engineering degree students at the appropriate level.

The challenge will be in four parts, each part will be awarded separate prizes

1. Work in teams to design, build and test a vehicle to compete with other teams.
2. Produce a poster to publicize the teams' work. The poster is a demonstration of the team's ability to sell their design solution.
3. Give a short presentation explaining the design and development of the device to demonstrate their verbal and presentational skills.
4. Submit a peer review where all students have the chance to vote for the best engineered solution to the problem.

The budget limit is **AED250** which has to be taken by the team itself. Submit the price list which will be verified by the judges during the competition.

In the spirit of the competition, it is expected that the device be designed, developed and manufactured by students within the facilities of their university. Any member of the team should have a good understanding of the design principles, theories, manufacturing methods and materials used.

This year the challenge is to design, which should be capable of undertaking a repeatable climbing movement externally to a commercial pipe fixed vertically, with an increasing load attached to it. The device must be totally self-contained and the method of starting it must be attached to, and travel with, the device at all times. The 'device' can be of any construction and propulsion method limited only by cost and size within the specification detailed below. Teams will compete head-to-head in heats for a place in the Regional final. The relative performance will be determined on the quickest times taken for the device to pull an increasing load to the top of their tube and return to the starting point at the bottom. The increasing load will be in the form of a chain, where the weight of chain to be lifted is specified as 0.15 kg/m.

The competing device can be based on any moving technology with contact with the vertical pipe, and it should:

- be started manually and perform the mission autonomously, without any other external support or control.
- be capable of performing on a real installation without requiring any post treatment or adjustment. Errors in vertical – plumb line may be present and the substrate of the commercial 22mm copper pipe will be as in purchased condition;
- start, return and / or stop on specific positions;
- engage with the top barrier;
- lift a changing load – delivered by a standardised chain;
- disengage the chain retention (clip)

Where relevant, see "*IMechE Design Challenge-General Specification 2026*"

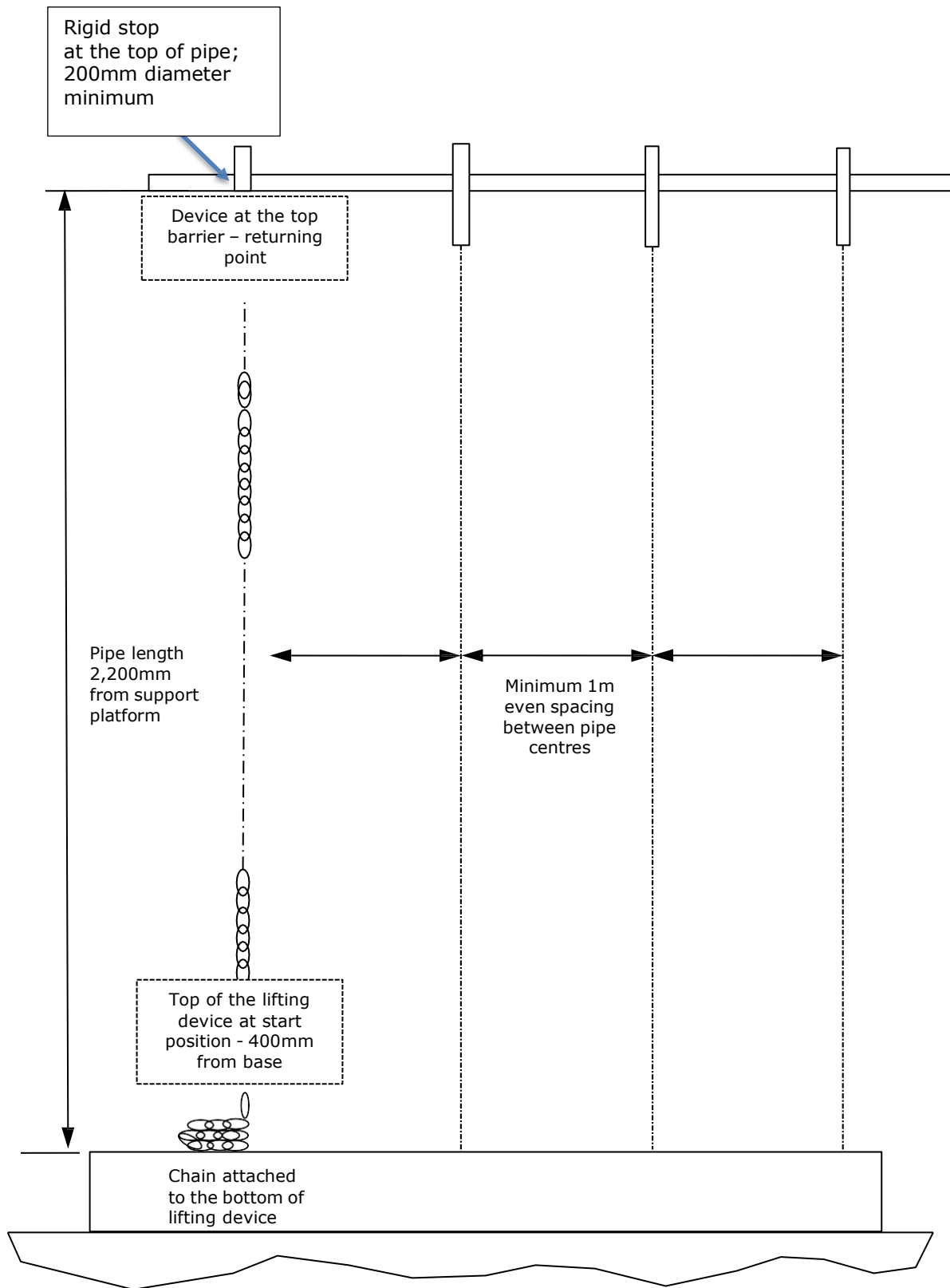


Figure 1 – Rig and device schematic

2. Rules

1. Each team will produce one single device.
2. The device must remain self-contained within a given envelope at any time of the inspection and during the competition.
3. Remote controllers, wireless technology, wireless programming and the use of GPS tracking is not allowed.
4. No proprietary and pre-programmed control units or devices may be used. Examples, but not limited to, are Lego Technic modules, vision camera – scanning systems for industrial applications, or similar.
5. Devices should be manufactured using available facilities and materials, using processes that students can undertake themselves. No manufacturing is to be outsourced.
6. Off the Shelf (OtS) components may be purchased, such as motors, batteries, gears, bearings, fasteners and similar.
7. All devices must be 'signed off' by the University, College or institute staff member the team belongs to. Clear details of the name and contact for the member shall be provided
8. The University, College or institute staff member(s) signing off the documents is(are) responsible for overall safety of the device.
9. All devices must be available for scrutineering prior to commencement of the competition
10. On the day of the competition, evidence of passing the scrutineering process is a IMechE sticker on the device and on the checklist associated. Vehicles without a sticker will not be allowed to compete (refer Section 3 for Device specification)
11. Photographic and filming activities may take place to monitor the activities in the competition lanes so that if there is a dispute it can easily be resolved.
12. Costs, mass, calculations and numbers in general will be presented together with the Units of measurements that identifies them.

3. Device Specification and Challenge Guidelines

1. The device can be of any type, but it must be completely self-contained and fit within a working envelope of 400mm diameter 300mm length maximum, excluding the automatic clip – see below *Figure 2*.

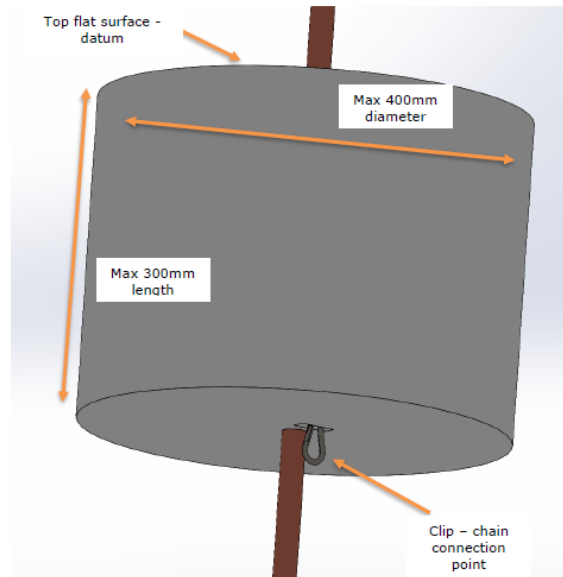


Figure 2. Maximum envelope, pipe and standard connecting clip

2. The device must always fit within the maximum working envelope during competition.
3. The device should have a visible green light, that remains lit for the duration of the run.
4. An additional visible red light and a sound (buzzer or similar) shall be on during some elements of the competition
5. The aim of this competition is to design a device that could perform the mission described below in less than 3 minutes in total.
6. Position the device with the top datum surface between the starting target lines as in Figure 3.
7. After the judges says "Go!"; Switch on the device – make the device operative and green light visible;
8. The device will start moving upwards, hitting the top barrier with the top datum surface;
9. It must remain in contact with the barrier up to 15 seconds, it should produce a red light and sound (buzzer);
10. Leave the top barrier, red light and sound switches off – only green light visible, return towards the starting point – first target;
11. Between the intermediate target (*Figure 4*) and the starting point – first target release the automatic clip and the chain connected to it,

12. The device shall include a single 50mm automatic clip – carabineer similar to www.screwfix.com 334FE as attachment of the load chain (Figure 5). This shall be included in BoM and considered as part of the vehicle, for both mass and cost considerations. The clip can protrude outside the maximum envelope, see Figure 2.
13. Progress movements towards the starting point – first target, time count is stopped at top surface passing it;
14. Stop autonomously or by contacting the ground without any operator intervention to complete the challenge;

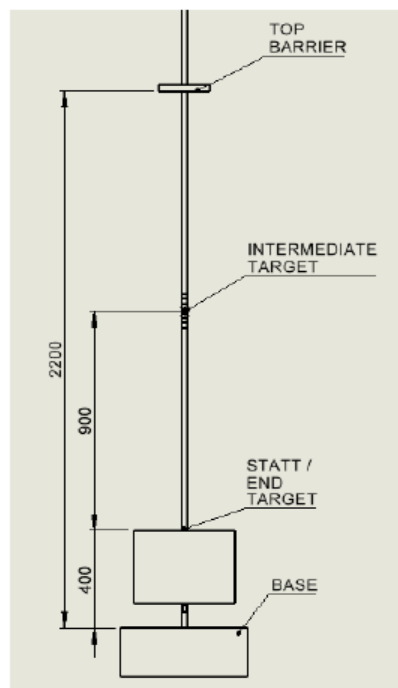


Figure 3. Single pipe rig schematic (for reference only)

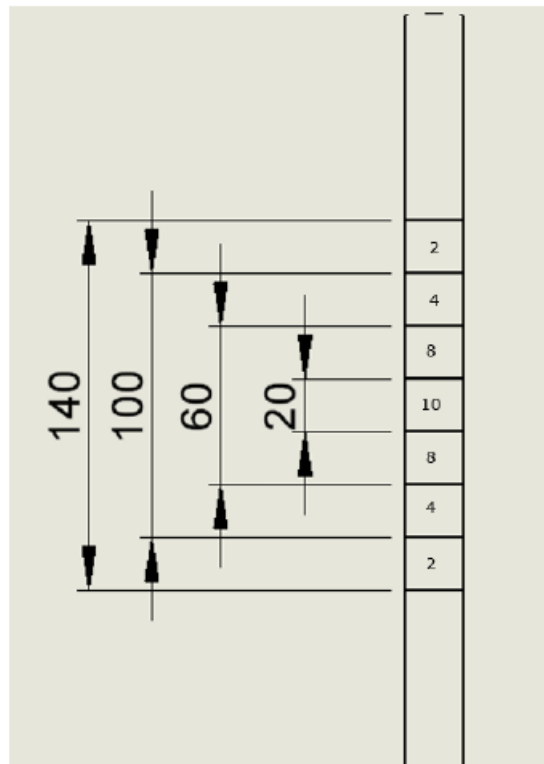


Figure 4. Intermediate target



Figure 5. Automatic clip – carabineer

Scoring - single movement competition:

15. The maximum score for the fastest time per run is 30 points. The slowest team in the heats or final will be awarded 10 points. Other teams will be awarded points proportionally.
16. Bonus points for holding the position before the start without operator support: 5 points
17. Bonus points for engagement with top barrier by emitting sound and visible light: 1 point for an engagement of 2 seconds or less, 5 points for engagement of 3 to 9

seconds, 10 points for engagement of 10 seconds or more, 0 points for engagement longer than 15 seconds;

18. Bonus points for releasing chain after passing the intermediate target (Figure 4) and before reaching the starting point – first target: 10 points
19. Bonus points are awarded only if the device continue or attempts to continue the mission to the following step.
20. A total of maximum 55 points is available per heat or run.
21. Points will be normalised to 100 for final scoring summary.

4. Target Competitors

First- and Second-Year Undergraduate Mechanical Engineering Students.

5. Number of participating teams from each University

Maximum 2 teams (Each team 2-5 members)

6. Prizes

- a. Cash prizes AED 1500 will be awarded to the winning team in the Final.
- b. There will be separate, independent, prizes for the poster, and peer review elements of the challenge.
- c. The winning team's institution shall be awarded a rolling trophy
- d. Appropriate certificates of recognition will be awarded to all students and the winning institute.

APPENDIX A –

Run	Assessment criterion	Unit	Points / scoring basis	Team 1	Team 2	Team 3	Team 4	Team 5
1st Run	Overall run time	s	Record actual time.					
	Duration of engagement	s	Record actual time.					
	Engagement score	pt	1 pt if 2 s or less; 5 pt if 3-9 s; 10 pt if 10-15 s; 0 pt if more than 15 s.					
	Net run time	s	Record actual time.					
	Net run time score	pt	Award according to competition scoring rules.					
	Holding position without operator	pt	Bonus: 5 pt.					
	Releasing the chain between mid and end in the descend	pt	Max 10 pt.					
	Run total	pt	Total points for the run.					
2nd Run	Overall run time	s	Record actual time.					
	Duration of engagement	s	Record actual time.					
	Engagement score	pt	1 pt if 2 s or less; 5 pt if 3-9 s; 10 pt if 10-15 s; 0 pt if more than 15 s.					
	Net run time	s	Record actual time.					
	Net run time score	pt	Award according to competition scoring rules.					
	Holding position without operator	pt	Bonus: 5 pt.					
	Releasing the chain between mid and end in the descend	pt	Max 10 pt.					
	Run total	pt	Total points for the run.					
3rd Run	Overall run time	s	Record actual time.					
	Duration of engagement	s	Record actual time.					
	Engagement score	pt	1 pt if 2 s or less; 5 pt if 3-9 s; 10 pt if 10-15 s; 0 pt if more than 15 s.					
	Net run time	s	Record actual time.					
	Net run time score	pt	Award according to competition scoring rules.					
	Holding position without operator	pt	Bonus: 5 pt.					
	Releasing the chain between mid and end in the descend	pt	Max 10 pt.					
	Run total	pt	Total points for the run.					
Final Summary	Total	pt	Final total after penalties.					
	Ranking	-	1 = highest rank.					

Final Record of Performance Template

	Total Score				
GROUP	HEAT 1	HEAT 2	HEAT 3	Highest Score of the three heats	RANKING
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

FINAL	FINAL TIME	RANKING
TBC		
TBC		
TBC		
TBC		

